

Qatar-1b

R-1791

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_150527 · created 2026-08-03T00:41:26+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457169.89879 +/- 0.00054 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.011
Transit depth (Rp/Rs)^2	3.13 +/- 0.38 [%]
Orbital Inclination (inc)	83.66 +/- 0.58
Airmass coefficient 1 (a1)	1.065 +/- 0.014
Airmass coefficient 2 (a2)	-0.046 +/- 0.022
Scatter in the residuals of the lightcurve fit is	1.29 %
Best Comparison Star	#3 - [55, 368]
Optimal Aperture	3.4
Optimal Annulus	4.04
Transit Duration (day)	0.0694 +/- 0.0034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.177 $\pm$ 0.011 vs 0.1463 (deviation 2.79 σ)
Duration fit vs published	0.0694 d vs 0.0692 d (deviation 0.07 σ)
Mid-time O-C	-0.6 min
Depth SNR	16.1
Residual scatter	1.29 %

Light curve

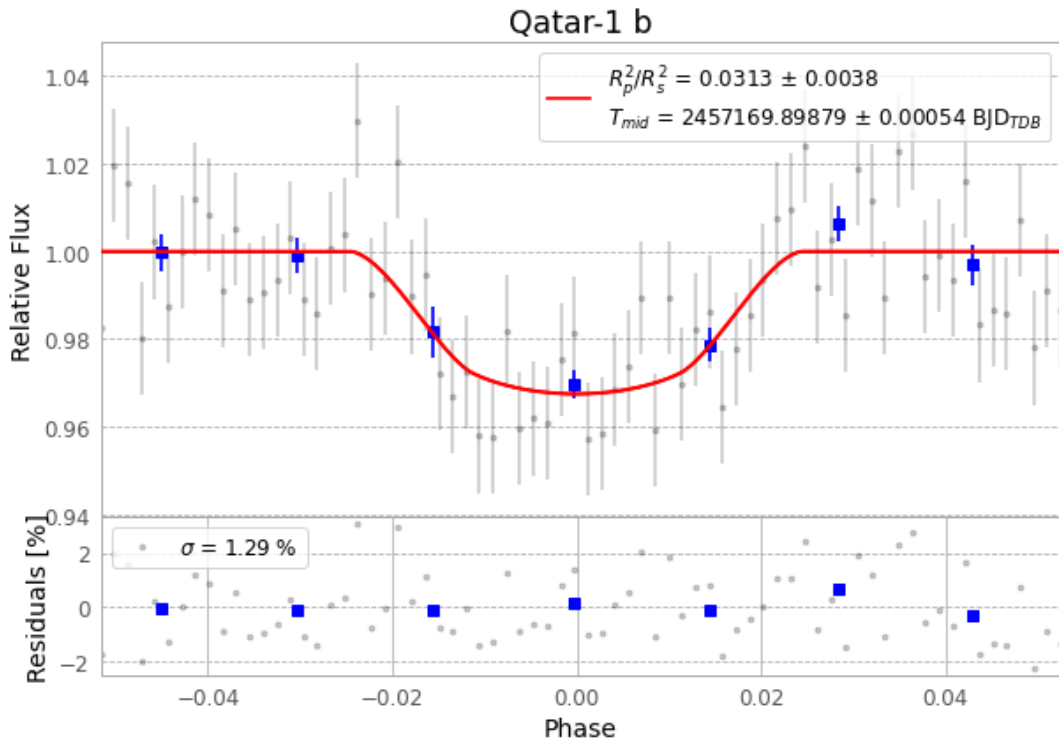


Figure 1 — FinalLightCurve_Qatar-1 b_2015-05-27.png (SHA-256 d1fd983c03761c76...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [323, 274], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 7b6713380b34f345...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

72 FITS frames (47 MB), Qatar-1150527074812.FITS → Qatar-1150527112112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-150527.log (6490f034b259...), FinalLightCurve_Qatar-1 b_2015-05-27.png (d1fd983c0376...), FinalLightCurve_Qatar-1 b_2015-05-27.csv (4125e229dfb3...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 00:41Z.